

Boot & Recovery Cheat Sheet

Get a Linux server that won't boot back on its feet: GRUB, rescue modes, lost passwords, bad fstab lines.

Using it: boot order and the GRUB menu

```
firmware --> GRUB --> kernel + initramfs --> systemd --> target  ← each stage hands over to the next
BIOS/UEFI  menu      mounts the real /      PID 1      multi-user  ← target = like an old runlevel
Hold Shift (BIOS) or tap Esc (UEFI) right after power-on  ← Ubuntu hides the menu; RHEL shows it
Press e on the entry, edit, then Ctrl-x (or F10) to boot  ← for this one boot only; Esc = cancel
linux /boot/vmlinuz-6.8.0-45-generic root=UUID=... ro quiet systemd.unit=rescue.target
```

Add one of these at the end of the **linux** line (**Ctrl-e** jumps there). Stuck at a **grub>** prompt? Type **normal**.

Add to the linux line	Gives you
systemd.unit=rescue.target	Single-user: root shell, filesystems mounted, no network (single = same)
systemd.unit=emergency.target	Bare minimum: root shell, / mounted read-only , nothing else
init=/bin/bash	Shell with no password, no systemd, / read-only: for resets or a locked root*
nomodeset	No kernel graphics drivers: fixes a black screen after the kernel loads

Rescue vs emergency

Both need root's password	rescue.target	emergency.target
Root filesystem	Mounted read-write	Mounted read-only
Other fstab mounts / services	Mounted / only the basics, no network	Not mounted / none
Use it when	A service or the desktop breaks the boot	A disk or fstab problem breaks rescue too

Fixing it

```
# Lost root password
(GRUB: e, add init=/bin/bash, Ctrl-x)
mount -o remount,rw /  ← / is read-only
passwd root  ← or passwd mark
touch /.autorelabel  ← SELinux (RHEL) only
sync  ← write changes to disk
exec /sbin/init  ← boot on (or reboot -f)
```

```
# Bad /etc/fstab line (emergency mode)
mount -o remount,rw /  ← so you can edit
systemctl --failed  ← which mount failed?
blkid  ← the real UUIDs
vi /etc/fstab  ← fix UUID, add nofail, or # it
systemctl daemon-reload
mount -a  ← no output = OK, then reboot
```

* Ubuntu locks root, so rescue/emergency won't give a shell. RHEL's docs use **rd.break** + **chroot /sysroot** instead.

Making changes permanent

```
sudoedit /etc/default/grub  ← e.g. GRUB_TIMEOUT=5, GRUB_CMDLINE_LINUX_DEFAULT="quiet nomodeset"
sudo update-grub  ← Debian/Ubuntu: applies it
sudo grub2-mkconfig -o /boot/grub2/grub.cfg  ← RHEL/Fedora: applies it (for kernel options, use grubby)
sudo grubby --update-kernel=ALL --args="nomodeset"  ← RHEL: kernel options, all kernels (--remove-args undoes)
```

Kernels

Command	Does	Command	Does
uname -r	Running kernel version	sudo update-initramfs -u	Rebuilds initramfs (Debian)
dpkg -l 'linux-image*'	Installed kernels (Debian)	sudo dracut -f	Rebuilds initramfs (RHEL)
rpm -q kernel	Installed kernels (RHEL)	Boot an older kernel	GRUB: <i>Advanced options</i>

Reading the output: after a bad boot

```
$ journalctl -b -1 -p err  ← errors from the previous boot (-b 0 = this one); also systemctl --failed
Oct 03 09:12:44 web01 systemd[1]: Failed to mount mnt-data.mount - /mnt/data.  ← the cause
$ systemd-analyze blame  ← slowest units first; 1min 30s = a timeout
$ systemd-analyze critical-chain  ← what the boot actually waited on
```

Remember: before rebooting a remote server after kernel/fstab/network changes, make sure you have **console access**.